Lecture 2: Waves and Diffusions

2.1 Wave Equation

In this section, we consider the following wave equation

$$\boxed{u_{tt} - c^2 u_{xx} = 0, \qquad \text{for } -\infty < x < \infty.} \tag{2.1}$$

First we have

Theorem 2.1 *The general solution of the wave equation* (2.51) *is*

$$\boxed{u(x,t) = f(x+ct) + g(x-ct)} \tag{2.2}$$

where f and g are two arbitrary functions of a single variable.

Proof: The proof based on the following observation

$$u_{tt} - c^2 u_{xx} = \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right) u = 0. \tag{2.3}$$

If we let $v = u_t + cu_x$, we must have $v_t - cv_x = 0$. Thus we have two first-order equations

$$v_t - cv_x = 0 \tag{2.4}$$

and

$$u_t + cu_x = v. \tag{2.5}$$

These two first-order equations are equivalent to (2.51) itself. First from §1.6.1 in lecture 1, equation (2.4) has the solution $v(x,t) = h(x+ct)$, where h is any function. So we only need to solve the equation

$$u_t + cu_x = h(x+ct). \tag{2.6}$$

Let $w(x,t) := u(x,t) - f(x+ct)$, then $w(x,t)$ satisfies equation

$$w_t + cw_x = u_t + cu_x - cf'(x+ct) - cf'(x+ct) = h(x+ct) - 2cf'(x+ct).$$

Now we choose $f'(s) = h(s)/2c$. Hence $w(x,t)$ satisfies equation

$$w_t + cw_x = 0.$$

Again from §1.6.1 in the lecture 1, $w(x,t) = g(x-ct)$, where g is any function. Notice that h is any function, so f is any function, and the general solution of (2.51) is

$$u(x,t) = f(x+ct) + g(x-ct),$$

as asserted by the theorem. ■

Remark 2.1.1 *The wave equation has a nice simple geometry. There are two families of characteristic lines, $x \pm ct = constant$. The general solution is the sum of functions from two families. One, $g(x-ct)$, is a wave of arbitrary shape traveling to the right at speed c. The other, $f(x+ct)$, is another shape traveling to the left at speed c.*

2.2 Initial Value Problem

The initial-value problem is to solve the wave equation (2.51) with the initial conditions

$$\boxed{u(x,0)=\phi(x), \qquad u_t(x,0)=\psi(x),} \tag{2.7}$$

where ψ and ϕ are arbitrary functions of x. For this problem, we have

Theorem 2.2 *The solution of the initial-value problem* (2.51) *and* (2.52) *is*

$$\boxed{u(x,t)=\frac{1}{2}[\phi(x+ct)+\phi(x-ct)]+\frac{1}{2c}\int_{x-ct}^{x+ct}\psi(s)ds.} \tag{2.8}$$

Proof: From Theorem 2.1, $u(x,t)=f(x+ct)+g(x-ct)$. So setting $t=0$, we get

$$\phi(x)=u(x,0)=f(x)+g(x). \tag{2.9}$$

Next by the chain rule, we differentiate $u(x,t)$ with respect to t and put $t=0$ to get

$$\psi(x)=u_t(x,0)=cf'(x)-cg'(x).$$

Hence we have

$$\phi'=f'+g', \qquad \frac{1}{c}\psi=f'-g'.$$

It means

$$f'=\frac{1}{2}\left(\phi'+\frac{\psi}{c}\right) \qquad \text{and} \qquad g'=\frac{1}{2}\left(\phi'-\frac{\psi}{c}\right).$$

Integrating, we get

$$f(s)=\frac{1}{2}\phi(s)+\frac{1}{2c}\int_0^s\psi+A, \qquad \text{and} \qquad g(s)=\frac{1}{2}\phi(s)-\frac{1}{2c}\int_0^s\psi+B, \tag{2.10}$$

where A and B are constants. Because of (2.9), we have $A+B=0$. So

$$u(x,t)=f(x+ct)+g(x-ct)=\frac{1}{2}\phi(x+ct)+\frac{1}{2c}\int_0^{x+ct}\psi+\frac{1}{2}\phi(x-ct)-\frac{1}{2c}\int_0^{x-ct}\psi$$

This implies (2.53). ■

Remark 2.2.1 *This is the solution formula for the initial-value problem, due to d'Alembert in 1746.*

Exercises

1. Prove that for any constant $h>0$, the general solution of the equation

$$\frac{\partial}{\partial x}\left[\left(1-\frac{x}{h}\right)^2\frac{\partial u}{\partial x}\right]=\frac{1}{a^2}\left(1-\frac{x}{h}\right)^2\frac{\partial^2 u}{\partial t^2}$$

is

$$u(x,t)=\frac{F(x-at)+G(x+at)}{h-x},$$

where F and G are two arbitrary functions of a single variable. Then to solve the PDE with the initial condition

$$u(x,0) = \phi(x), \qquad \text{and} \qquad u_t(x,0) = \psi(x).$$

2. Solve $u_{tt} - c^2 u_{xx} = 0$, $u(x,0) = e^x$, $u_t(x,0) = \sin x$.

3. Solve $u_{tt} = c^2 u_{xx}$, $u(x,0) = \log(1+x^2)$, $u_t(x,0) = 4 + x$.

4. A *Spherical wave* is a solution of the three-dimensional wave equation of the form $u(r,t)$, where r is the distance to the origin (the spherical coordinate). The spherical wave equation takes the form

$$u_{tt} = c^2 \left(u_{rr} + \frac{2}{r} u_r \right).$$

(a) Prove that under the change of coordinate $r = \sqrt{x^2+y^2+z^2}$ and under the assumption that $u(x,y,z,t) = u(r,t)$, solutions of the three dimensional wave equation $u_{tt} - c^2(u_{xx} + u_{yy} + u_{zz}) = 0$ are the solutions of the spherical wave equation.

(b) Change variables $v = ru$ to get the equation for v: $v_{tt} = c^2 v_{rr}$.

(c) Solve for v and thereby solve the spherical wave equation.

(d) Solve the spherical wave equation with the initial conditions $u(r,0) = \phi(r)$ and $u_t(r,0) = \psi(r)$.

2.3 Important Properties of Solutions of the Wave Equation

2.3.1 Causality (finite speed of propagation)

As a corollary of Theorem 2.2, solutions of the wave equation (2.51), which governs the propagation of waves, satisfies the following *principle of causality.*

Corollary 2.3 *No part of the wave determined by the wave equation* (2.51) *goes faster than speed c.*

Proof: From Theorem 2.2,

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds. \tag{2.11}$$

The effect of an initial position $\phi(x)$ is a pair of waves traveling in either direction at speed c and at half the original amplitude. While the effect of an initial velocity $\psi(x)$ is a wave spreading out at speed $x \le c$ in both directions. So part of the wave may lag behind (if there is an initial velocity), but no part goes faster than speed c. ■

An initial condition (position or velocity or both) at the point $(x_0, 0)$ can affect the solution for $t > 0$ only in the sector $\{(x,t) \,|\, |x| \le x_0 + ct\}$, which is called the *domain of influence* of the point $(x_0, 0)$. In addition, the domain of influence of an interval $|x| \le R$ is a sector $|x| \le R + ct$.

There is also an 'inverse' way to express causality. Notice that the value of $u(x_0, t_0)$ only depends on the value of ϕ at the two points $x_0 \pm ct_0$, and it depends only on the values of ψ within the interval $[x_0 - ct_0, x_0 + ct_0]$. We therefore say that the interval $(x_0 - ct_0, x_0 + ct_0)$ is the *interval of dependence* of the point (x_0, t_0) on $t = 0$. We call the entire triangle with vertices at $(x_0 - ct_0, 0)$, $(x_0 + ct_0, 0)$ and (x_0, t_0) the *domain of dependence* or the *past history* of the point (x_0, t_0).

2.3.2 Energy

From physics, we know that the kinetic energy is $KE = \frac{1}{2}\int \rho u_t^2 dx$, and that the potential energy is $PE = \frac{1}{2}\int T u_x^2 dx$. Thus the total energy is

$$\boxed{E(t) := \frac{1}{2}\int_{-\infty}^{\infty} (\rho u_t^2 + T u_x^2) dx.} \tag{2.12}$$

Here we assume that u_t and u_x vanish at $x = \pm\infty$. Differentiating the total energy, we can pass the derivative under the integral sign to get

$$\frac{dE(t)}{dt} = \rho \int_{-\infty}^{\infty} u_t u_{tt} dx + T \int_{-\infty}^{\infty} u_x u_{tx} dx$$

Then we substitute the PDE $\rho u_{tt} = T u_{xx}$ and integrate by parts to get

$$\frac{dE(t)}{dt} = T \int_{-\infty}^{\infty} u_t u_{xx} dx + T \int_{-\infty}^{\infty} u_x u_{tx} dx = T u_t u_x \Big|_{-\infty}^{\infty} - T \int_{-\infty}^{\infty} u_{xt} u_x dx + T \int_{-\infty}^{\infty} u_x u_{tx} dx = 0.$$

Thus $E(t)$ is a constant independent of t. This is the law of *conservation of energy.*

2.4 Well-Posedness of the Initial Problem of the Wave Equation

In this section, we consider the following initial problem with a source $f(x,t)$

$$\boxed{\begin{aligned} &u_{tt} - c^2 u_{xx} = f \qquad \text{for } t > 0, \\ &u = \phi(x), \quad \text{and} \quad u_t = \psi(x) \qquad \text{when } t = 0. \end{aligned}}$$

2.4.1 Uniqueness

From the law of conservation of energy, we can easily obtain the following uniqueness result.

Theorem 2.4 *There is at most one solution of the initial problem.*

Proof: Assume there are two solutions u and v of the initial problem. Let $w = u - v$. Then

$$w_{tt} - c^2 w_{xx} = u_{tt} - c^2 u_{xx} - (v_{tt} - c^2 v_{xx}) = 0, \qquad \text{for } t > 0,$$

and

$$w(x,0) = u(x,0) - v(x,0) = 0, \qquad w_t(x,0) = u_t(x,0) - v_t(x,0) = 0.$$

So

$$w_x = w_t = 0 \qquad \text{when } t = 0.$$

Hence the total energy for w at $t = 0$ vanish. Notice that the total energy for w is constant due to the homogeneous wave equation which w satisfies. Then

$$E(t) = E(0) = 0.$$

So $u - v = w = 0$ for $t > 0$. ■

2.4.2 Existence

Theorem 2.5 *The unique solution of the initial problem is*

$$\boxed{u(x,t)=\frac{1}{2}[\phi(x+ct)+\phi(x-ct)]+\frac{1}{2c}\int_{x-ct}^{x+ct}\psi+\frac{1}{2c}\int_0^t\int_{x-c(t-s)}^{x+c(t-s)}f(y,s)dyds.} \tag{2.13}$$

The proof of this theorem is based on the following observation.

Lemma 2.6 (Duhamel's principle) *If $w(x,t;\tau)$ is a solution to the problem*

$$\begin{cases} w_{tt}-c^2w_{xx}=0 & (t>\tau),\\ w=0,\ w_t=f(x,\tau) & (t=\tau)\end{cases} \tag{2.14}$$

then

$$u(x,t)=\int_0^t w(x,t;\tau)d\tau \tag{2.15}$$

is a solution of the problem

$$\begin{cases} u_{tt}-c^2u_{xx}=f(x,t) & (t>0),\\ u=0,\ u_t=0 & (t=0).\end{cases} \tag{2.16}$$

Proof: Let $t'=t-\tau$, then w satisfies

$$\begin{cases} w_{t't'}-c^2w_{xx}=0 & (t'>0),\\ w=0,\ w_{t'}=f(x,\tau) & (t'=0)\end{cases}$$

So by theorem 2.2,

$$w(x,t;\tau)=\frac{1}{2c}\int_{x-ct'}^{x+ct'}f(\xi,\tau)d\xi=\frac{1}{2c}\int_{x-c(t-\tau)}^{x+c(t-\tau)}f(\xi,\tau)d\xi$$

So

$$u(x,t)=\frac{1}{2c}\int_0^t\int_{x-c(t-\tau)}^{x+c(t-\tau)}f(\xi,\tau)d\xi d\tau. \tag{2.17}$$

Now taking derivatives on u, we have

$$\begin{aligned} u_t=&\frac{1}{2c}\int_{x-c(t-t)}^{x+c(t-t)}f(\xi,t)d\xi+\frac{1}{2}\int_0^t f(x+c(t-\tau),\tau)d\tau+\frac{1}{2}\int_0^t f(x-c(t-\tau),\tau)d\tau\\ =&\frac{1}{2}\int_0^t[f(x+c(t-\tau),\tau)+f(x-c(t-\tau),\tau)]d\tau\\ u_{tt}=&f(x,t)+\frac{c}{2}\int_0^t[\frac{\partial f}{\partial x}(x+c(t-\tau),\tau)-\frac{\partial f}{\partial x}(x-c(t-\tau),\tau)]d\tau\\ u_x=&\frac{1}{2c}\int_0^t[f(x+c(t-\tau),\tau)-f(x-c(t-\tau),\tau)]d\tau\\ u_{xx}=&\frac{1}{2c}\int_0^t[\frac{\partial f}{\partial x}(x+c(t-\tau),\tau)-\frac{\partial f}{\partial x}(x-c(t-\tau),\tau)]d\tau. \end{aligned}$$

Obviously, $u_{tt}-c^2u_{xx}=f(x,t)$, $u(x,0)=0$, and $u_t(x,0)=0$. ■

Remark 2.6.1 *Duhamel principle can be connected to the Newton's law in the following sense. The physical meaning of the source term f in the wave equation is a external force. In a small time interval $d\tau$, the total external force is approximately $f(x,\tau)d\tau$. By the Newton's law, the speed generated by the external force is $f(x,\tau)d\tau$. We treat it as the initial speed at time τ. Thus we solve the problem* (2.14) *to have a solution $w(x,t;\tau)$. Obviously as the effect of the external force f in the whole time, the solution of problem* (2.16) *is the integral of w with respect to the small time interval τ.*

The proof of theorem 2.5 is now a direct corollary based on the Duhamel principle. Actually, we can divided the initial problem into problem (2.16) and the problem in theorem 2.2. Since the wave equation is linear, the solution of the initial problem is the summation of the solution in theorem 2.2 and the solution in the Duhamel principle which is given by (2.17).

2.4.3 Stability

To make statement precise, we need a way to measure the 'nearness' of functions. We will illustrate this concept by using the *uniform norms*:

$$||u|| = \max_{-\infty<x<\infty} |u(x)|$$

and

$$||u||_T = \max_{-\infty<x<\infty, 0\le t\le T} |u(x,t)|.$$

Here T is fixed. Suppose that $u_(x,t)$ is the solution with data $(\phi_1(x), \psi_1(x), f_1(x,t))$ and $u_2(x,t)$ is the solution with data $(\phi_2(x), \psi_2(x), f_2(x,t))$. Let $u = u_1 - u_2$. Then by (2.13),

$$u(x,t) = \frac{1}{2}[\phi_1(x+ct)-\phi_2(x+ct)+\phi_1(x-ct)-\phi_2(x-ct)]+\frac{1}{2c}\int_{x-ct}^{x+ct}(\psi_1-\psi_2)+\frac{1}{2c}\int_0^t\int_{x-c(t-s)}^{x+c(t-s)}(f_1-f_2)dyds.$$

Hence

$$|u(x,t)| \le \max|\phi_1-\phi_2| + t\cdot\max|\psi_1-\psi_2| + \frac{t^2}{2}\cdot\max|f_1-f_2|.$$

So

$$||u_1-u_2||_T \le ||\phi_1-\phi_2|| + T||\psi_1-\psi_2|| + \frac{T^2}{2}||f_1-f_2||_T.$$

Thus for any small number ϵ, if $||\phi_1-\phi_2|| < \delta$, $||\psi_1-\psi_2|| < \delta$, and $||f_1-f_2|| < \delta$, and if $(1+T+T^2)\delta \le \epsilon$, then

$$||u_1-u_2||_T < (1+T+T^2)\delta \le \epsilon.$$

Exercises

1. Show that the wave equation has the following invariance properties.

   (a) Any translate $u(x-y,t)$, where y is fixed, is also a solution.

   (b) Any derivative, say u_x, of a solution is also a solution.

   (c) The dilated function $u(ax,at)$ is also a solution.

2. Solve $u_{tt} - c^2u_{xx} = xt$, $u(x,0) = 0$, $u_t(x,0) = 0$.

3. Solve $u_{tt} - c^2u_{xx} = e^{at}$, $u(x,0) = 0$, $u_t(x,0) = 0$.

4. Solve $u_{tt} - c^2u_{xx} = \cos x$, $u(x,0) = \sin x$, $u_t(x,0) = 1+x$.

2.5 Waves in Higher Dimension

2.5.1 Three dimension case

Theorem 2.7 *The solution to the initial value problem*

$$\boxed{\begin{cases} u_{tt} - c^2(u_{xx} + u_{yy} + u_{zz}) = 0 \\ u(x,y,z,0) = \phi(x,y,z), \qquad u_t(x,y,z,0) = \psi(x,y,z). \end{cases}} \tag{2.18}$$

is

$$\boxed{u(x_0,y_0,z_0,t_0) = \frac{1}{4\pi c^2 t_0}\iint_S \psi(x,y,z)dS + \frac{\partial}{\partial t_0}\left[\frac{1}{4\pi c^2 t_0}\iint_S \phi(x,y,z)dS\right],} \tag{2.19}$$

where S is the sphere of center (x_0,y_0,z_0) and radius ct_0.

Remark 2.7.1 *The famous formula* (2.55) *is due to Poisson but is known as Kirchhoff's formula.*

Remark 2.7.2 (Huygens's principle.) *According to* (2.55)*, the value of $u(x_0,y_0,z_0,t_0)$ depends only on the values of ϕ and ψ on the spherical surface $S = \{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = c^2t_0^2\}$, but ont on the value of ϕ and ψ inside this sphere. It means that* any solution of the three-dimensional wave equation propagates at *exactly* the speed c, no faster and no slower.

Proof: *Step 1.* Let

$$\bar{u}(x_0,y_0,z_0,r,t) := \frac{1}{4\pi r^2}\iint_{S_r} u(x,y,z,t)dS,$$

where S_r is the sphere of center (x_0,y_0,z_0) and radius r. First by the change of coordinate,

$$\bar{u}(x_0,y_0,z_0,r,t) = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} u(x_0+ra_1, y_0+ra_2, z_0+ra_3, t)\sin\theta d\theta dw,$$

where (x,y,z) are expressed in terms of the spherical coordinates (r,θ,w). Here a_i are functions of (θ,w) and satisfy $a_1^2+a_2^2+a_3^2 = 1$. Hence

$$u(x_0,y_0,z_0,t) = \lim_{r\to 0}\bar{u}(x_0,y_0,z_0,r,t).$$

Step 2. Now we are going to derive the equation which $\bar{u}$ satisfies. Notice that by the chain rule and the divergent theorem,

$$\bar{u}_r = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\sum_{i=1}^{3}\frac{\partial u}{\partial x_i}a_i\sin\theta d\theta dw = \frac{1}{4\pi r^2}\iint_{S_r}\sum_{i=1}^{3}\frac{\partial u}{\partial x_i}a_i dS = \frac{1}{4\pi r^2}\iint_{D_r}\triangle_{(x,y,z)}u dxdydz,$$

where $(x_1,x_2,x_3) = (x,y,z)$ and D_r is the ball with boundary surface S_r. Then

$$\bar{u}_{rr} = -\frac{1}{2\pi r^3}\iint_{D_r}\triangle_{(x,y,z)}u dxdydz + \frac{1}{4\pi r^2}\iint_{S_r}\triangle_{(x,y,z)}u dS.$$

So

$$\bar{u}_{tt} - c^2(\bar{u}_{rr} + \frac{2}{r}\bar{u}_r) = \frac{1}{4\pi r^2}\iint_{S_r}(u_{tt} - c^2\triangle_{(x,y,z)}u)dS = 0.$$

Notice that all the calculation above is valid for $r > 0$. Define $\bar{u}(x_0, y_0, z_0, -r, t) = \bar{u}(x_0, y_0, z_0, r, t)$, then it is easy to check that $\bar{u}$ satisfies the equation above for all $-\infty < r < \infty$.

Let $v = r\bar{u}$. Then easily v satisfies the following one-dimensional wave equation

$$v_{tt} - c^2 v_{rr} = 0. \tag{2.20}$$

For the initial conditions, obviously,

$$v\big|_{t=0} = r\bar{u}\big|_{t=0} = r\bar{\phi}(x_0, y_0, z_0, r) := \frac{1}{4\pi r}\iint_{S_r} \phi(x, y, z)dS, \tag{2.21}$$

$$v_t\big|_{t=0} = r\bar{u}_t\big|_{t=0} = r\bar{\psi}(x_0, y_0, z_0, r) := \frac{1}{4\pi r}\iint_{S_r} \psi(x, y, z)dS. \tag{2.22}$$

Here for $r > 0$, we define $-r\bar{\phi}(x_0, y_0, z_0, -r) = -r\bar{\phi}(x_0, y_0, z_0, r)$, and $-r\bar{\psi}(x_0, y_0, z_0, -r) = -r\bar{\psi}(x_0, y_0, z_0, r)$.

Step 3. From the d'Alembert formula (2.53),

$$v(x_0, y_0, z_0, r, t) = \frac{1}{2}[(r + ct)\bar{\phi}(x_0, y_0, z_0, r + ct) + (r - ct)\bar{\phi}(x_0, y_0, z_0, r - ct)] + \frac{1}{2c}\int_{r-ct}^{r+ct} s\bar{\psi}(x_0, y_0, z_0, s)ds.$$

Hence by the L'Hôpital's rule,

$$\begin{aligned} u(x_0, y_0, z_0, t) &= \lim_{r\to 0} \bar{u}(x_0, y_0, z_0, r, t) = \lim_{r\to 0} \frac{1}{r} v(x_0, y_0, z_0, r, t) \\ &= \lim_{r\to 0} \frac{1}{2r}[(r + ct)\bar{\phi}(x_0, y_0, z_0, r + ct) + (r - ct)\bar{\phi}(x_0, y_0, z_0, r - ct)] + \frac{1}{2cr}\int_{r-ct}^{r+ct} s\bar{\psi}(x_0, y_0, z_0, s)ds \\ &= \frac{\partial}{\partial t}[t\bar{\phi}(x_0, y_0, z_0, ct)] + t\bar{\psi}(x_0, y_0, z_0, ct) \end{aligned}$$

It completes the proof. ■

2.5.2 Two dimension case

Theorem 2.8 *The solution to the initial value problem*

$$\boxed{\begin{cases} u_{tt} - c^2(u_{xx} + u_{yy}) = 0 \\ u(x, y, 0) = \phi(x, y), \qquad u_t(x, y, 0) = \psi(x, y). \end{cases}} \tag{2.23}$$

is

$$\boxed{u(x_0, y_0, t_0) = \frac{1}{2\pi c}\left[\frac{\partial}{\partial t_0}\int_0^{ct_0}\int_0^{2\pi} \frac{\phi(x_0 + r\cos\theta, y_0 + r\sin\theta)}{\sqrt{(ct_0)^2 - r^2}} r d\theta dr + \int_0^{ct_0}\int_0^{2\pi} \frac{\psi(x_0 + r\cos\theta, y_0 + r\sin\theta)}{\sqrt{(ct_0)^2 - r^2}} r d\theta dr\right].} \tag{2.24}$$

Remark 2.8.1 *The formula* (2.57) *shows that the value of* $u(x_0, y_0, t_0)$ *does depend on the values of* ϕ *and* ψ inside *the cone:* $(x - x_0)^2 + (y - y_0)^2 \le c^2 t_0^2$. *It means that* Huygens's principle is false in two dimensions.

Proof: The key idea is to regard $u(x, y, t)$ as a solution of the three-dimensional problem *which just happens not to depend on* z. So it must be given by the Kirchhoff formula.

For the sake of simplicity, we assume that $\phi = 0$ and $(x_0, y_0) = (0, 0)$. Hence by (2.55),

$$u(0,0,t) = \frac{1}{4\pi c^2 t} \iint_{x^2+y^2+z^2=c^2t^2} \psi(x,y) dS.$$

Notice that it is twice integral over the top hemisphere $z = \sqrt{c^2t^2 - x^2 - y^2}$. Moreover, $dxdy = dS \cdot \cos\gamma$, where

$$\cos\gamma = \frac{\sqrt{(ct)^2 - x^2 - y^2}}{ct}$$

Thus

$$u(0,0,t) = \frac{2}{4\pi c^2 t} \iint_{x^2+y^2\le c^2t^2} \frac{ct\psi(x,y)}{\sqrt{(ct)^2 - x^2 - y^2}} dxdy = \frac{1}{2\pi c} \int_0^{ct} \int_0^{2\pi} \frac{\psi(r\cos\theta, r\sin\theta)}{\sqrt{(ct)^2 - r^2}} r d\theta dr.$$

The argument to the case that $\phi \neq 0$ and $(x_0, y_0) \neq (0, 0)$ is similar. ■

Exercises

1. To solve the wave equation in three dimensions with the initial data $\phi = 0$ and $\psi(x, y, z) = y$.

2. To solve the wave equation in two dimensions with the initial data $\phi = x^2(x + y)$ and $\psi = 0$.

3. Why doesn't the method of spherical means (the method to get formula (2.55)) work for two-dimensional waves?

4. To deduce the d'Alembert formula (2.53) by regarding $u(x, t)$ as a solution of the two-dimensional problem.

2.6 Fourier Transformation

In this section, we will introduce an integral transformation named *Fourier transformation*, which is powerful to solve linear partial differential equations.

Definition 2.9 *Let $f(x)$ is absolutely integrable in $\mathbb{R}$. The function $g(\lambda) = F[f](\lambda)$ defined by*

$$\boxed{F[f](\lambda) := \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx,} \tag{2.25}$$

where $x \in \mathbb{R}$, is called Fourier transformation *of f, and the function $f(x) = F^{-1}[g](x)$ defined by*

$$\boxed{F^{-1}[g](x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} g(\lambda) e^{i\lambda x} d\lambda,} \tag{2.26}$$

is called inverse Fourier transformation *of g, provided the integrals on the right hand side exist.*

Remark 2.9.1 *The definition of the inverse Fourier transformation comes from the following fact:*

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda \int_{-\infty}^{\infty} f(\xi) e^{i\lambda(x-\xi)} d\xi. \tag{2.27}$$

Obviously, we have

Proposition 2.10 *The Fourier transformation is linear, i.e., for any constants α, β, and functions f_1, f_2, we have*

$$\boxed{F[\alpha f_1 + \beta f_2] = \alpha F[f_1] + \beta F[f_2].} \tag{2.28}$$

In order to study the Fourier transformation of multiplication, we need to introduce the notion of *convolution*.

Definition 2.11 *For two given functions $f_1(x)$ and $f_2(x)$, if for $x \in \mathbb{R}$, the integral*

$$\boxed{f(x) = f_1 \star f_2 := \int_{-\infty}^{\infty} f_1(x-t) f_2(t) dt} \tag{2.29}$$

exists, then we call $f(x)$ is the convolution of functions $f_1(x)$ and $f_2(x)$.

Obviously, $f_1 \star f_2 = f_2 \star f_1$, i.e., the convolution is commutative. Then we have the following important property.

Proposition 2.12 *The Fourier transformation of a convolution of two functions equals the multiplication of the Fourier transformation of the functions, i.e.,*

$$\boxed{F[f_1 \star f_2] = F[f_1] \cdot F[f_2].} \tag{2.30}$$

Proof: By definition,

$$\begin{aligned} F[f_1 \star f_2] =& \int_{-\infty}^{\infty} e^{-i\lambda x} dx \int_{-\infty}^{\infty} f_1(x-t) f_2(t) dt \\ =& \int_{-\infty}^{\infty} f_2(t) dt \int_{-\infty}^{\infty} f_1(x-t) e^{-i\lambda x} dx \\ =& \int_{-\infty}^{\infty} f_2(t) dt \int_{-\infty}^{\infty} f_1(\xi) e^{-i\lambda(t+\xi)} d\xi \\ =& \int_{-\infty}^{\infty} f_2(t) e^{-i\lambda t} dt \int_{-\infty}^{\infty} f_1(\xi) e^{-i\lambda \xi} d\xi \\ =& F[f_1] \cdot F[f_2]. \end{aligned}$$

■

Similarly, we also have the following property.

Proposition 2.13 *For functions $f_1(x)$ and $f_2(x)$, we have*

$$\boxed{F[f_1 \cdot f_2] = \frac{1}{2\pi} F[f_1] \star F[f_2].} \tag{2.31}$$

The next property sates that the Fourier transformation transforms a differential equation into an algebraic equation.

Proposition 2.14 *If the Fourier transformations of $f(x)$ and $f'(x)$ exist, and if $f(x) \to 0$ when $|x| \to \infty$, then*

$$\boxed{F[f'(x)] = i\lambda F[f(x)].} \tag{2.32}$$

Proof: By integrating by part,

$$\begin{aligned} F[f'(x)] &= \int_{-\infty}^{\infty} f'(x)e^{-i\lambda x}dx \\ &= [f(x)e^{-i\lambda x}]|_{x=-\infty}^{x=\infty} + \int_{-\infty}^{\infty} i\lambda f(x)e^{-i\lambda e}dx \\ &= i\lambda \int_{-\infty}^{\infty} f(x)e^{-i\lambda x}dx = i\lambda F[f(x)]. \end{aligned}$$

■

Similarly, by integrating by part, we have

Proposition 2.15 *If both the Fourier transformation of $f(x)$ and $xf(x)$ exist, then*

$$\boxed{F[-ixf(x)] = \frac{d}{d\lambda}F[f].} \tag{2.33}$$

Remark 2.15.1 *For the multi-dimensional case, we can define the Fourier transformation similarly as*

$$F[f] := \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \cdots, x_n)e^{-i(x_1\lambda_1 + \cdots + x_n\lambda_n)}dx_1 \cdots dx_n. \tag{2.34}$$

And the transformation has all similar properties as in Proposition 2.10–2.15.

2.7 Initial Value Problem to the Heat Equation

We use the Fourier transformation to solve the problem

$$\boxed{\begin{cases} u_t = a^2 u_{xx} & (-\infty < x < \infty, 0 < t < \infty) \\ u(x,0) = \phi(x). \end{cases}} \tag{2.35}$$

For fixed t, let

$$\hat{u}(\lambda, t) := F[u(x,t)], \qquad \hat{\phi}(\lambda) := F[\phi(x)].$$

By proposition 2.10 and proposition 2.14, we have the following ordinary equation

$$\frac{d\hat{u}}{dt} = -a^2\lambda^2\hat{u},$$

with the initial condition

$$\hat{u}(\lambda, 0) = \hat{\phi}(\lambda).$$

The solution to the ordinary equation is

$$\hat{u}(\lambda, t) = \hat{\phi}(\lambda)e^{-a^2\lambda^2 t}. \tag{2.36}$$

By proposition 2.12, we have

$$u(x,t) = F^{-1}[\hat{\phi}(\lambda)e^{-a^2\lambda^2 t}] = \phi \star F^{-1}[e^{-a^2\lambda^2 t}]. \tag{2.37}$$

Notice that

$$\begin{aligned} F^{-1}[e^{-a^2\lambda^2 t}] =& \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-a^2\lambda^2 t + i\lambda x} d\lambda \\ =& \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-a^2 t\left(\lambda - \frac{ix}{2a^2 t}\right)^2} d\lambda \cdot e^{-\frac{x^2}{4a^2 t}}, \end{aligned}$$

and that

$$\int_{-\infty}^{\infty} e^{-a^2 t\left(\lambda - \frac{ix}{2a^2 t}\right)^2} d\lambda = \int_{-\infty}^{\infty} e^{-a^2 t\lambda^2} d\lambda = \frac{1}{a\sqrt{t}}\int_{-\infty}^{\infty} e^{-y^2} dy = \frac{\sqrt{\pi}}{a\sqrt{t}}.$$

Here we use the fact that $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$. It can be proved by the following identity:

$$\left(\int_0^{\infty} e^{-x^2} dx\right)^2 = \int_0^{\infty} e^{-x^2} dx \cdot \int_0^{\infty} e^{-y^2} dy = \int_0^{\infty} e^{-r^2} r \int_0^{\frac{\pi}{2}} d\theta dr = \frac{\pi}{4}.$$

Thus we have

$$F^{-1}[e^{-a^2\lambda^2 t}] = \frac{1}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}} := K(x,t), \tag{2.38}$$

and the solution to the initial value problem (2.35) is

$$\boxed{u(x,t) = \frac{1}{2a\sqrt{\pi t}}\int_{-\infty}^{\infty} \phi(\xi) e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi.} \tag{2.39}$$

Remark 2.15.2 *Formula* (2.39) *is called* Poisson's formula *and the function* $K(x,t)$ *defined by* (2.38) *is called* heat kernel *or* fundamental solution *of the heat equation.*

Corollary 2.16 *The speed of propagation of heat flows is infinite.*

Proof: Notice that from (2.39), the value of $u(x,t)$ depends on the values of the initial data $\phi(y)$ for *all* y, where $-\infty < y < \infty$. Conversely, the value of ϕ at a point x_0 has an *immediate effect everywhere* (for $t > 0$). Therefore the speed of propagation is infinite. ■

Based on (2.39), we can derive the solution to the initial value problem of the heat equation with a source term by the Duhamel's principle.

Theorem 2.17 *The solution to the initial value problem*

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = f(x,t) & (-\infty < x < \infty, 0 < t < \infty) \\ u(x,0) = \phi(x). \end{cases}} \tag{2.40}$$

is

$$\boxed{u(x,t) = \frac{1}{2a\sqrt{\pi t}}\int_{-\infty}^{\infty} \phi(\xi) e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \frac{1}{2a\sqrt{\pi}}\int_0^t \int_{-\infty}^{\infty} \frac{f(\xi,\tau)}{\sqrt{t-\tau}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi d\tau.} \tag{2.41}$$

Proof: Since the heat equation is linear, we only need to consider the problem

$$\begin{cases} u_t - a^2 u_{xx} = f(x,t) & (-\infty < x < \infty, 0 < t < \infty) \\ u(x,0) = 0. \end{cases} \tag{2.42}$$

By the Duhamel's principle, the solution of this problem is

$$u(x,t) = \int_0^t w(x,t;\tau)d\tau, \tag{2.43}$$

where $w = w(x,t;\tau)$ is a solution of the following problem

$$\begin{cases} w_t - a^2 w_{xx} = 0 & (-\infty < x < \infty, \tau < t < \infty) \\ w(x,\tau) = f(x,\tau). \end{cases} \tag{2.44}$$

Then by the Poisson formula (2.35), it is easy to see that the solution to the problem (2.42) is

$$u(x,t) = \frac{1}{2a\sqrt{\pi}} \int_0^t \int_{-\infty}^{\infty} \frac{f(\xi,\tau)}{\sqrt{t-\tau}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi d\tau. \tag{2.45}$$

■

Exercises

1. To find the Fourier transformation of $e^{-\eta x^2}$, for any constant $\eta > 0$.
2. Solve the heat equation without source with the initial condition $u(x,0) = e^{-x}$.
3. Solve the heat equation without source with the initial condition $u(x,0) = e^{3x}$.
4. To prove that if $u_1(x,t)$ and $u_2(x,t)$ are solutions of the problem

$$\begin{cases} \frac{\partial u_i}{\partial t} - a^2 \frac{\partial^2 u_i}{\partial x^2} = 0 & (-\infty < x < \infty, 0 < t < \infty) \\ u_i(x,0) = \varphi_i(x), \end{cases}$$

respectively, then $u(x,y,t) = u_1(x,t)u_2(y,t)$ is the solution of the problem

$$\begin{cases} u_t - a^2(u_{xx} + u_{yy}) = 0 & (-\infty < x,y < \infty, 0 < t < \infty) \\ u(x,0) = \varphi_1(x)\varphi_2(y). \end{cases} \tag{2.46}$$

2.8 Important Properties of Solutions of the Heat Equation

2.8.1 Maximum principle

Theorem 2.18 *If $u(x,t)$ satisfies the heat equation $u_t - a^2 u_{xx} = 0$ in a rectangle R in space-time, i.e., in $R := \{(x,t) \big| 0 \le x \le l, 0 \le t \le T$, then the maximum value of $u(x,t)$ is attained either initially ($t = 0$) or on the lateral sides ($x = 0$ or $x = l$).*

Proof: Let M be the maximum value of $u(x,t)$ on the three sides $t = 0$, $x = 0$, and $x = l$. So we only need to show that $u(x,t) \le M$ in R.

Let ϵ be a positive constant and let $v(x,t) = u(x,t) + \epsilon x^2$. Now in the rectangle R,

$$v_t - a^2 v_{xx} = u_t - a^2 u_{xx} - a^2(\epsilon x^2)_{xx} = -2a^2\epsilon < 0.$$

Meanwhile, if $v(x,t)$ attains its maximum value at an *interior* point (x_0,t_0), then $v_t(x_0,t_0) = 0$ and $v_{xx}(x_0,t_0) \le 0$. It is a contradiction. Assume now that $v(x,t)$ attains its maximum value at a point

(x_0, t_0) on the *top* edge, i.e., $t_0 = T$. Then $v_x(x_0, t_0) = 0$ and $v_{xx}(x_0, t_0) \le 0$ as before. furthermore, since $v(x_0, t_0) > v(x_0, t_0 - \delta)$, we have

$$v_t(x_0, t_0) = \lim_{\delta \to 0+} \frac{v(x_0, t_0) - v(x_0, t_0 - \delta)}{\delta} \ge 0.$$

We can again reach a contradiction. Thus we have

$$v(x, t) \le M + \epsilon l^2.$$

So

$$u(x, t) \le M + \epsilon(l^2 - x^2),$$

for any constant $\epsilon > 0$. Therefore

$$u(x, t) \le M \qquad \text{in } R.$$

■

2.8.2 Energy

Let us consider the problem

$$\boxed{\begin{cases} u_t = a^2 u_{xx} & (-\infty < x < \infty, 0 < t < \infty) \\ u(x, 0) = \phi(x). \end{cases}} \tag{2.47}$$

Define the energy $E(t)$ as

$$E(t) := \int_{-\infty}^{\infty} [u(x, t)]^2 dx. \tag{2.48}$$

Here we assume that u and u_x vanish at $x = \pm\infty$. Differentiating the energy, we can pass the derivative under the integral sign to get

$$\frac{dE(t)}{dt} = \int_{-\infty}^{\infty} u u_t dx.$$

Then we substitute the PDE $u_t = a^2 u_{xx}$ and integrate by parts to get

$$\frac{dE(t)}{dt} = a^2 \int_{-\infty}^{\infty} u u_{xx} dx = a^2 u u_x|_{-\infty}^{\infty} - a^2 \int_{-\infty}^{\infty} [u_x(x, t)]^2 dx = -a^2 \int_{-\infty}^{\infty} [u_x(x, t)]^2 dx < 0.$$

Thus $E(t)$ is decreasing. So

$$\int_{-\infty}^{\infty} [u(x, t)]^2 dx \le \int_{-\infty}^{\infty} [u(x, 0)]^2 dx. \tag{2.49}$$

2.8.3 Well-posedness

In this subsection, we consider the well-posedness of the following initial value problem

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = 0 & \text{for } t > 0, \\ u = \phi(x), & \text{when } t = 0. \end{cases}}$$

Obviously, the existence is guaranteed by the Poisson formula (2.39). Therefore we only need to consider the uniqueness and stability of the initial value problem. Actually we have

Theorem 2.19 *The solution to the initial value problem above is unique and stable with respect to the initial data, provided that the solution is uniformly bounded by a constant B, i.e., $|u(x,t)| \le B$ for all $t \ge 0$ and $-\infty < x < \infty$.*

Proof: We will use the maximum principle to prove it.

First we prove the *uniqueness*. Assume that there are two solutions u_1 and u_2 with the same initial data $\phi(x)$. Let $w = u_1 - u_2$. Then $|w(x,t)| \le 2B$ for all $t \ge 0$ and $-\infty < x < \infty$. And w satisfies the following problem

$$\boxed{\begin{cases} w_t - a^2 w_{xx} = 0 & \text{for } t > 0, \\ w = 0, & \text{when } t = 0. \end{cases}}$$

Obviously, we only need to prove $w \equiv 0$. For any fixed point (x_0, t_0) with $t_0 > 0$, consider the rectangle

$$R_L : 0 \le t \le t_0, \quad |x - x_0| \le L,$$

where L is a positive constant. Let

$$v(x,t) := \frac{4B}{L^2}\left(\frac{(x-x_0)^2}{2} + a^2 t\right).$$

Obviously, $v_t - a^2 v_{xx} = 0$ in R_L. $v(x,0) \ge 0 = w(x,0)$, and $v(x_0 \pm L, t) \ge 2B \ge w(x_0 \pm L, t)$. Hence from the maximum principle, in R_L, $v(x,t) \ge w(x,t)$, which means

$$w(x,t) \le \frac{4B}{L^2}\left(\frac{(x-x_0)^2}{2} + a^2 t\right).$$

Similarly, we can prove that in R_L, $-v(x,t) \le u(x,t)$, which means

$$w(x,t) \ge -\frac{4B}{L^2}\left(\frac{(x-x_0)^2}{2} + a^2 t\right).$$

So if $u(x,t)$ takes value at the special point (x_0, t_0), then

$$|w(x_0,t_0)| \le \frac{4B}{L^2} a^2 t_0.$$

Let $L \to \infty$, then

$$w(x_0, t_0) = 0.$$

Thus we proved the uniqueness.

Second, for the stability, we assume that there are two solutions. One is u_1 with the initial data ϕ_1, while the other one is u_2 with the initial data ϕ_2. Let $w = u_1 - u_2$. Then $|w(x,t)| \le 2B$ for all $t \ge 0$ and $-\infty < x < \infty$. And w satisfies the following problem

$$\boxed{\begin{cases} w_t - a^2 w_{xx} = 0 & \text{for } t > 0, \\ w = \phi_1 - \phi_2, & \text{when } t = 0. \end{cases}}$$

Obviously, we only need to prove $|w(x,t)| \le \max_{x\in\mathbb{R}} |\phi_1(x) - \phi_2(x)|$. For any fixed point (x_0, t_0) with $t_0 > 0$, let

$$v(x,t) := \frac{4B}{L^2}\left(\frac{(x-x_0)^2}{2} + a^2 t\right) + \max_{x\in\mathbb{R}} |\phi_1(x) - \phi_2(x)|.$$

Then similar to the proof of the uniqueness, from the maximum principle, we have

$$|w(x_0,t_0)| \le \frac{4B}{L^2}a^2t_0 + \max_{x\in\mathbb{R}}|\phi_1(x)-\phi_2(x)|.$$

Hence let $L\to\infty$, then

$$|w(x_0,t_0)| \le \max_{x\in\mathbb{R}}|\phi_1(x)-\phi_2(x)|.$$

Thus we proved the stability. ■

2.9 Comparison of Waves and Diffusions

To conclude this lecture, the fundamental properties of these two equations are summarized in the following table.

Property	Waves	Diffusions
(i) Speed of propagation?	Finite ($\le c$)	infinite
(ii) Well-posed for $t>0$?	Yes	Yes
(iii) Maximum principle	No	Yes
(iv) Behavior at $t\to\infty$?	Energy is constant so does not decay	Decay to zero
(v) Information	Transported	Lost gradually

Please check the precedent section for more details.

Exercises.

1. To prove the uniqueness and stability of the following initial-boundary value problem:

$$\begin{cases} u_t - a^2u_{xx} = 0, & \text{for } 0<x<L \text{ and } t>0 \\ u(x,0)=\phi(x), & \\ u(0,t)=g(t), \qquad u(l,t)=h(t), & \end{cases} \tag{2.50}$$

where h, g, and ϕ are given smooth functions.

2. Consider the solution $1-x^2-2a^2t$ of the heat equation. Find the locations of its maximum and its minimum in the closed rectangle $(0\le x\le l,\ 0\le t\le T)$.

3. Consider a solution of the diffusion equation $u_t-u_{xx}=0$ in $\{0\le x\le l,\ 0\le t\le\infty\}$.

   (a) Let $M(T)$ be the maximum of $u(x,t)$ in the rectangle $(0\le x\le l,\ 0\le t\le T)$. Does $M(T)$ increase or decrease as a function of T?

   (b) Let $m(T)$ be the minimum of $u(x,t)$ in the rectangle $(0\le x\le l,\ 0\le t\le T)$. Does $m(T)$ increase or decrease as a function of T?

4. Consider the diffusion equation $u_t-u_{xx}=0$ in $\{0<x<1,\ 0<t<\infty\}$ with $u(0,t)=u(1,t)=0$ and $u(x,0)=4x(1-x)$.

   (a) Show that $0<u(x,t)<1$ for all $t>0$ and $0<x<1$.

   (b) Show that $u(x,t)=u(1-x,t)$ for all $t\ge0$ and $0\le x\le1$.

   (c) Use the energy to show that $\int_0^1 u^2dx$ is a strictly decreasing function of t.

5. Show that there is no maximum principle for the wave equation. (Hint: take $\phi=0$ and $\psi\neq0$.)

2.10 Appendix

Theorem 2.20 *The solution of the wave equation*

$$u_{tt} - c^2 u_{xx} = f(x,t), \qquad for\ -\infty < x < \infty, \tag{2.51}$$

with the initial conditions

$$u(x,0) = \phi(x), \qquad u_t(x,0) = \psi(x), \tag{2.52}$$

is

$$\boxed{u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.} \tag{2.53}$$

Theorem 2.21 *The solution to the initial value problem*

$$\begin{cases} u_{tt} - c^2(u_{xx} + u_{yy} + u_{zz}) = 0 \\ u(x,y,z,0) = \phi(x,y,z), \qquad u_t(x,y,z,0) = \psi(x,y,z). \end{cases} \tag{2.54}$$

is

$$\boxed{u(x_0,y_0,z_0,t_0) = \frac{1}{4\pi c^2 t_0}\iint_S \psi(x,y,z)dS + \frac{\partial}{\partial t_0}\left[\frac{1}{4\pi c^2 t_0}\iint_S \phi(x,y,z)dS\right],} \tag{2.55}$$

where S is the sphere of center (x_0,y_0,z_0) and radius ct_0.

Theorem 2.22 *The solution to the initial value problem*

$$\begin{cases} u_{tt} - c^2(u_{xx} + u_{yy}) = 0 \\ u(x,y,0) = \phi(x,y), \qquad u_t(x,y,0) = \psi(x,y). \end{cases} \tag{2.56}$$

is

$$\boxed{u(x_0,y_0,t_0) = \frac{1}{2\pi c}\left[\frac{\partial}{\partial t_0}\int_0^{ct_0}\int_0^{2\pi} \frac{\phi(x_0 + r\cos\theta, y_0 + r\sin\theta)}{\sqrt{(ct_0)^2 - r^2}} r d\theta dr + \int_0^{ct_0}\int_0^{2\pi} \frac{\psi(x_0 + r\cos\theta, y_0 + r\sin\theta)}{\sqrt{(ct_0)^2 - r^2}} r d\theta dr\right].} \tag{2.57}$$

Definition 2.23 *Let $f(x)$ is absolutely integrable in $\mathbb{R}$. The function $g(\lambda) = F[f](\lambda)$ defined by*

$$\boxed{F[f](\lambda) := \int_{-\infty}^{\infty} f(x)e^{-i\lambda x}dx,} \tag{2.58}$$

where $x \in \mathbb{R}$, is called Fourier transformation *of f.*

Theorem 2.24 *The solution to the initial value problem*

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = f(x,t) \qquad (-\infty < x < \infty, 0 < t < \infty) \\ u(x,0) = \phi(x). \end{cases}} \tag{2.59}$$

is

$$\boxed{u(x,t) = \frac{1}{2a\sqrt{\pi t}}\int_{-\infty}^{\infty} \phi(\xi)e^{-\frac{(x-\xi)^2}{4a^2 t}}d\xi + \frac{1}{2a\sqrt{\pi}}\int_0^t \int_{-\infty}^{\infty} \frac{f(\xi,\tau)}{\sqrt{t-\tau}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}}d\xi d\tau.} \tag{2.60}$$